Selection Sort Dry Run

arr : [4, 6, 9, 7, 5]
$\quad\quad$ 0 $\;$ 1 $\;$ 2 $\;$ 3 $\;$ 4

for i = 0 & j = 1
min_ind = i
as arr[min_ind] < arr[j]
arr[0] < arr[1]
4 < 6
j++ (i = 0, j = 2)
as arr[min_ind] < arr[j]
4 < 9
j++ (i = 0, j = 3)
as arr[min_ind] < arr[j]
4 < 7
j++ (i = 0, j = 4)
as arr[min_ind] < arr[j]
4 < 5
exit inner loop
[4 $\quad$ 6, 9, 7, 5]
sorted $\quad$ unsorted

~~for~~ i++
(i = 1, j = 2)
min_ind = 1
as arr[1] ~~<~~ arr[2]
6 ~~<~~ 9
~~min_ind =~~ j++ (i = 1, j = 3)
as arr[1] < arr[3]
6 < 7
j++ (i = 1, j = 4)

graph.addEdge(1,3);

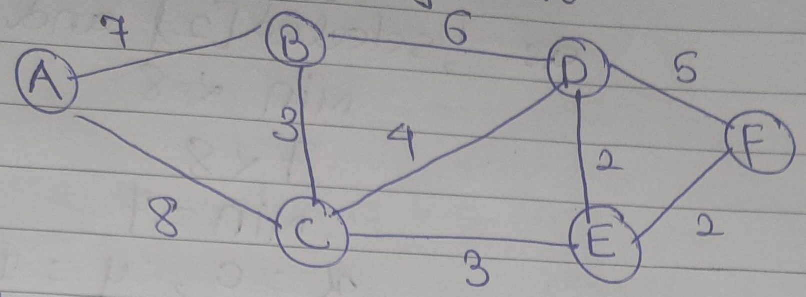
as arr[1] > arr[4] arr[4, 6, 9, 7, 5]
6 > 5
min_ind = i (1)
exit inner loop
swap arr[min_ind], arr[i] arr[4, 5, 9, 7, 6]
i++

(i = 2, j = 3)
as arr[2] > arr[3]
min_ind = 3
j++ (i = 2, j = 4)
as arr[3] > arr[4]
min_ind = 4
exit inner loop
swap arr[min_ind], arr[i] arr[4, 5, 6, 7, 9]

i++
(i = 3, j = 4)
as arr[3] < arr[4]
7 < 9
skip
exit inner loop
exit outer loop

sorted array: [4, 5, 6, 7, 9]

Prim's Algorithm Dry Run



Graph representation:

	A	B	C	D	E	F
A	0	7	8	0	0	0
B	7	0	3	6	0	0
C	8	3	0	4	3	0
D	0	6	4	0	2	5
E	0	0	3	2	0	2
F	0	0	0	5	2	0

labels = [A, B, C, D, E, F] , V = 6

selected = [T, F, F, F, F, F]

edges = 0

tot_cost = 0

min = INT32   x = 0, y = 0

for i in range (v)

(for i = 0)

selected [0] = T

for j = 0

as ! selected[0] and graph[0][0] = 0 :

skip

for j = 1

as ! selected [1] and g[0][1] = 7

min > 7

∴ min = 7

x = 0, y = 1

for j = 2

as !selected[2] and g[0][2] = 8

min < 8

7 < 8

∴ min = 7

x = 0, y = 1

for j = 3

g[0][3] = 0

j = 4 g[0][4] = 0 → skip

j = 5 g[0][5] = 0

∴ tot_cost = 7

edges = 1, selected[1] = T

A - B = 7

i++ (i = 1)

for (j = 0)

selected[0] = T but g[1][0] = $\cancel{7}$, skip

j++

for (j = 1)

selected[1] = T but g[1][1] = 0 skip

j++

for (j = 2)

selected[2] = F and g[1][2] = 3 ~~skip~~

as min > 3

∴ min = 3

x = 1, y = 2

j++

for (j = 3)

selected[3] = F and g[1][3] = 6

as 3 > 6 = F

skip

j++

for (j = 4)

selected [4] = f , g[1][4] = 0 , skip

for(j = 5) selected [5] = f , g[1][5] = 0 , skip

∴ tot_cost = 7 + 3 = 10

edges = 2 . selected [2] = T .

B - C = 3

i++ (i = 2)

For (j = 0)

selected [0] = T , g[2][0] = 8 , skip

j++

For (j = 1)

selected [1] = T g[2][1] = 3 , skip

j++

for (j = 2)

selected [2] = T g[2][2] = 0 , skip

j++

For (j = 3)

selected [3] = F and g[2][3] = 4

as min > 4

∴ min = 4

∴ x = 2 , y = 3

j++

For (j = 4)

selected[4] = f and g[2][4] = 3

as 4 > 3

∴ min = 3

∴ x = 2 , y = 4

j++

for (j = 5)
selected[5] = F but g[2][5] = 0 skip

∴ tot_cost = 7+3+3 = 13
edges = 3, selected[4] = T (x = 2, y = 4)
C - E : 3

i++
for (i = 3)
selected[3] = F
skip

i++
for (i = 4)
selected[4] = T
for (j = 0), selected[0] = T, j++
selected[1] = T → skip, j++
selected[2] = T → skip, j++
for (j = 3)
selected[3] = F g[4][3] = 2
as min ≤ 2
∴ min = 2
x = 4, y = 3 selected[3] = T
j++
for (j = 4)
selected[4] = T, g[4][4] = 0, skip
j++
for (j = 5)
selected[5] = F, g[4][5] = 2, ~~skip~~
as 2 > 2 = F
∴ min = 2
x = 4, y = 3 selected[5] = T

∴ tot_cost = 7 + 3 + 3 + 2 = 15.
edges = 4, selected [3] = T ($x = 4, y = 3$)
E - D : 2

i++
For (i = 5)
selected [5] = T.
for (j = 0)
selected [0] = T skip, j++
for j = 1, selected [1] = T skip, j++
for j = 2 selected [2] = T skip, j++
for j = 3 selected [3] = T skip, j++
for (j = 4)
selected [4] = T, g[5][4] = 2
min > 2
∴ min = 2
$x = 5$, $y = 4$
exit outer loop i
∴ tot_cost = 7 + 3 + 3 + 2 + 2 = 17
edges = 5 = $\overset{(V-1)}{(6-1)}$
F - E = 2

∴ A - B : 7
B - C : 3
C - E : 3
E - D : 2
E - F : 2